Dennis Sciama, the third great mentor of the era, had yet another style. He devoted the 1960s and early 1970s almost exclusively to providing an optimal environment for his Cambridge University students to grow in. Because he relegated his own personal research and career to second place, after those of his students, he was never promoted to the august position of "Professor" at Cambridge (a position much higher than being a professor in America). It was his students, far more than he, who reaped the rewards and the kudos. By the end of the 1970s two of his former students, Stephen Hawking and Martin Rees, were Cambridge Professors.

Sciama was a catalyst; he kept his students closely in touch with the most important new developments in physics, worldwide. Whenever an interesting discovery was published, he would assign a student to read and report on it to the others. Whenever an interesting lecture was scheduled in London, he would take or send his entourage of students down on the train to hear it. He had exquisitely good sense about what ideas were interesting, what issues were worth pursuing, what one should read in order to get started on any research project, and whom one should go to for technical advice.

Sciama was driven by a desperate desire to know how the Universe is made. He himself described this drive as a sort of metaphysical angst. The Universe seemed so crazy, bizarre, and fantastic that the only way to deal with it was to try to understand it, and the best way to understand it was through his students. By having his students solve the most challenging problems, he could move more quickly from issue to issue than if he paused to try to solve them himself.

Black Holes Have No Hair

Among the discoveries of the golden age, one of the greatest was that "a black hole has no hair." (The meaning of this phrase will become clear gradually in the coming pages.) Some discoveries in science are made quickly, by individuals; others emerge slowly, as a result of diverse contributions from many researchers. The hairlessness of black holes was of